DAW - WEB INTERFACE DESIGN

# UNIT 2. ACCESSIBLE WEB DESIGN

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# REVIEW OF MARKUP LANGUAGE PROGRAMMING

**Objectives:**

- Analyze and select appropriate colors and typefaces for on-screen visualization
- Use frames and tables to present information in an organized way
- Identify new elements and tags introduced in the latest version of HTML, HTML5
- Recognize the possibilities for modifying HTML tags
- Evaluate the usefulness of style sheets (CSS) to achieve a uniform design across the entire website

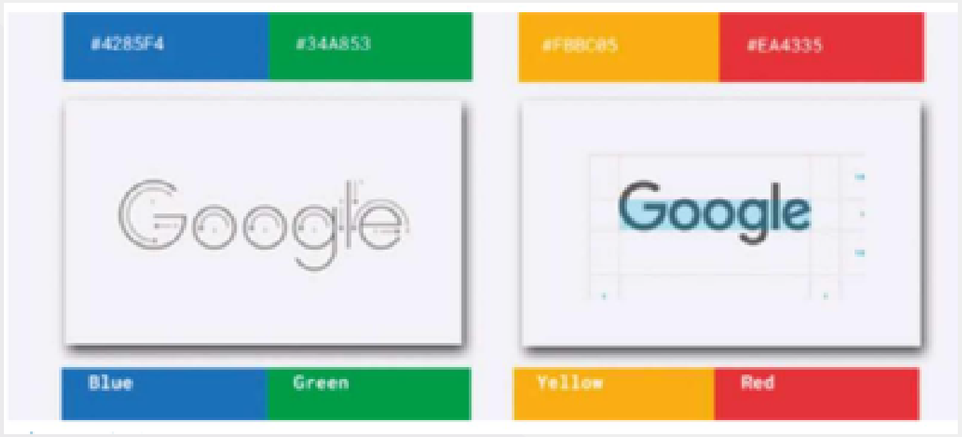
# STYLE GUIDES. MAIN ELEMENTS AND DESIGN GUIDELINES

- Just as in daily life we follow traffic rules or action protocols, web design also relies on what are known as style guides
- Standardizing styles through the creation of a style guide, which gathers the criteria and rules a project must follow, ensures that a webpage or website has a more uniform and appealing appearance for the user
- Just as style guides are created for magazines or books, they are also used in the development of web interfaces
- This set of rules for the design and writing of documents is common across all types of media, both spoken and visual
- Their goal is to achieve a **coherent, effective and correct message**

# AUDIENCE AND PURPOSE OF A STYLE GUIDE

- A style guide is intended for the designers and programmers responsible for creating the web interface
- Its mission is to support better decision-making both during the design process and in the later maintenance phase of the website
- This is especially important because website maintenance is often carried out by multiple people
- The guide should include aspects such as: how photographs and images are introduced, the color palette, typefaces, and other visual and structural criteria

# AUDIENCE AND PURPOSE OF A STYLE GUIDE (II)



# TYPOGRAPHY

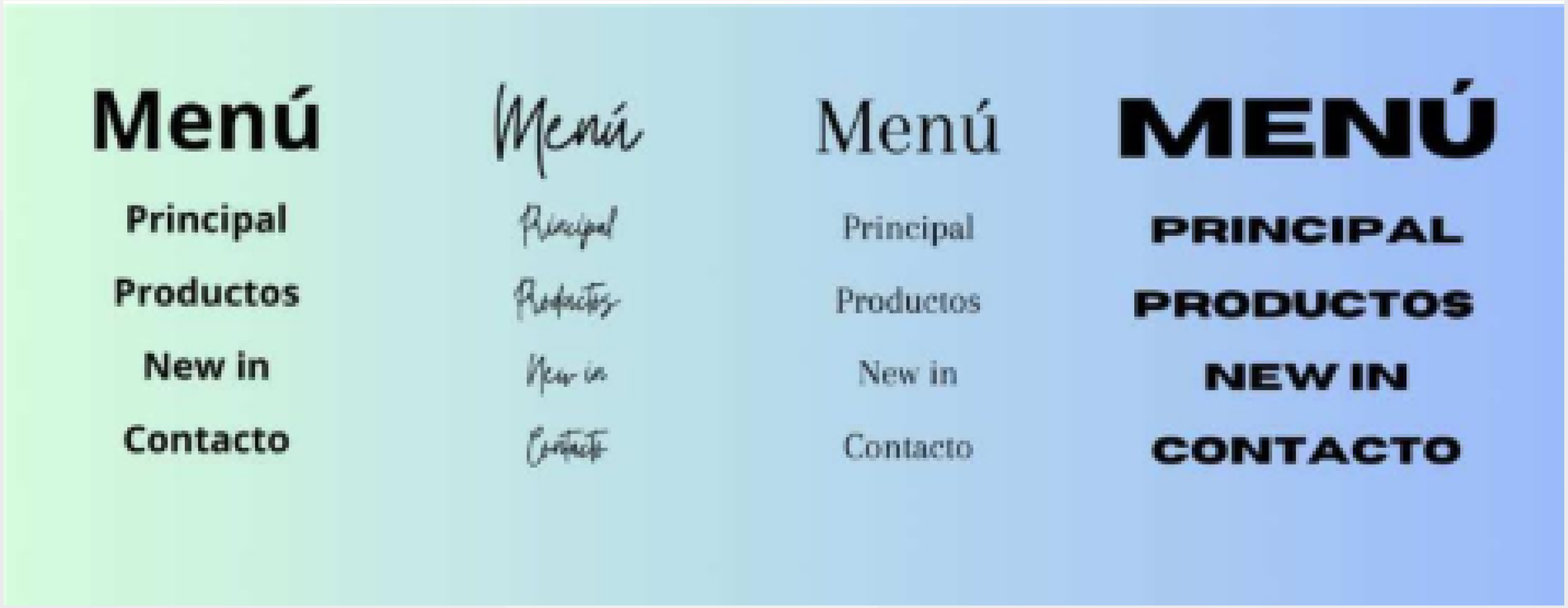
- Refers to the typeface chosen for a design.
- Familiar fonts improve readability and user experience.
- Select functional fonts that display correctly across all browsers.
- Good typography improves readability, usability, and accessibility.

# CHOOSING THE RIGHT TYPOGRAPHY

When selecting a typeface, consider:

- **Font family:** Choose widely supported fonts (e.g., Arial) to ensure compatibility across browsers.
- **Style:** Avoid overusing bold or underlining. Underlined text is commonly associated with hyperlinks and may confuse users.
- **Font size:** Use a size that is easy to read on different screens and devices.
- **Color:** Dark text on a light background provides better readability. Avoid low-contrast or risky combinations, such as pink text on a red background.

# CHOOSING THE RIGHT TYPOGRAPHY (II)



# FROM SAFE FONTS TO WEB FONTS

- Early websites relied on safe fonts, which were pre-installed on users' devices.
- Today, most browsers support the **@font-face** rule.
- This allows designers to use almost any typeface through services such as **Google Fonts**.
- Users do not need to have the font installed on their device.
- TrueType (TTF) and OpenType (OTF) fonts can be converted into web formats such as:
  - WOFF (Web Open Font Format)
  - EOT (Embedded OpenType)

# TYPOGRAPHY BEST PRACTICES FOR WEB DESIGN

a) Use no more than 2-3 typefaces in the same project.
b) Choose clear and readable fonts.
c) Ensure good contrast between text and background.
d) Avoid overly long lines of text.
e) Use a line spacing (leading) 1.5 points larger than the body text size.
f) Avoid justified text; left alignment is usually easier to read.
g) Minimize the use of underlining and reserve it for hyperlinks.

TYPOGRAPHY SERIES

@kenny.kenray

# Font Psychology

CHOOSING THE RIGHT STYLE WITH PURPOSE

## Serif

Aa

**Baskerville / Garamond / Georgia**

**Reliable**
**Respectable**
**Dependable**
**Conventional**
**Neutral**

Books
Newspapers
Fashion

## Sans Serif

Aa

**Futura / Gill Sans / Montserrat**

**Clean**
**Simple**
**Futuristic**
**Modern**
**Direct**

Signages
Social Media
Websites

## Script

*Aa*

**Bickham Script / Snell Roundhand**

**Personal**
**Feminine**

Invitations
Greetings
Weddings

free font

## Slab Serif

Aa

**Archer / Museo Slab / Clarendon**

**Bold**
**Confident**
**Strong**
**Reliable**
**Solid**

Billboards
Sports
Displays

© Ken Ray Communications Sdn Bhd

Save for reference

Ps | DESIGN BASICS

FREE FONTS. BIG IMPACT.

# Free Fonts Every Designer Should Have

VERSATILE / PROFESSIONAL / 100% FREE / PERFECT FOR ANY PROJECT

**Great design starts with great fonts.**

Here are the best free fonts every designer should have in their toolkit — clean, versatile and ready for any project.

1 SANS-SERIF (CLEAN & VERSATILE)

**Inter**

Clean, modern and highly readable.
Perfect for UI, web and body text.

**Use for:** UI, web, body text — Google Fonts

2 SANS-SERIF (MODERN & STYLISH)

**Montserrat**

Geometric, clean and professional.
Works great for headlines and branding.

**Use for:** Headlines, branding, posters — Google Fonts

3 SERIF (CLASSIC & TIMELESS)

**Playfair Display**

Elegant and sophisticated.
Perfect for luxury, fashion and editorial design.

**Use for:** Titles, luxury, editorial — Google Fonts

4 SERIF (MODERN & CLEAN)

**Lora**

Balanced, readable and stylish.
Great for long-form content and print.

**Use for:** Body text, articles, blogs — Google Fonts

5 DISPLAY (BOLD & IMPACTFUL)

**Bebas Neue**

Strong, tall and attention-grabbing.
Perfect for posters, banners and headlines.

**Use for:** Headlines, posters, banners — Google Fonts

6 SCRIPT (ELEGANT & CREATIVE)

**Allura**

Stylish, flowing and personal.
Adds a human touch to your designs.

**Use for:** Quotes, logos, invitations — Google Fonts

7 MONO (TECH & MINIMAL)

**Space Grotesk**

Modern, clean and technical.
Great for tech, UI and minimal designs.

**Use for:** UI, tech, minimal — Google Fonts

8 DISPLAY (UNIQUE & CREATIVE)

**Anton**

Bold, confident and full of character.
Perfect for making a strong statement.

**Use for:** Posters, social media, branding — Google Fonts

9 SERIF (STRONG & CLASSY)

**Merriweather**

Readable, professional and timeless.
Great for long text and editorial work.

**Use for:** Articles, blogs, print — Google Fonts

10 SANS-SERIF (FRIENDLY & ROUND)

**Poppins**

Clean, modern and friendly.
Works well for both headings and body text.

**Use for:** UI, web, branding — Google Fonts

11 SERIF (ELEGANT & REFINED)

**Cormorant Garamond**

Graceful, refined and sophisticated.
Perfect for luxury and high-end brands.

**Use for:** Luxury, editorial, branding — Google Fonts

12 SCRIPT (CASUAL & NATURAL)

**Pacifico**

Friendly, casual and full of personality.
Great for creative and fun designs.

**Use for:** Logos, quotes, social media — Google Fonts

★ QUICK TIPS

- Use trusted sources like Google Fonts.
- Pair a display font with a clean sans-serif or serif.
- Aa — Keep a good balance (2–3 fonts per design).
- Make sure the font fits your brand's personality.
- Always check license (most are free for personal & commercial use).

*"The right font doesn't just show your words — it shows your personality."*

@gfx.om

BETTER FONTS / BETTER DESIGNS

TYPOGRAPHY RULES
GOOD TYPOGRAPHY MAKES CONTENT CLEAR, ATTRACTIVE & MEMORABLE
1. HIERARCHY
Use size, weight, and contrast to show importance and guide the reader's eye.
MOST IMPORTANT
LEAST IMPORTANT
Main Heading
Subheading
Section Title
This is body text. It supports the message and should be easy to read.
Learn more...
2. CONTRAST
Use contrast in font styles, sizes, and weights to create visual interest.
BOLD vs. light
LARGE vs. small
Serif vs. Sans-serif
UPPERCASE vs. lowercase
3. ALIGNMENT
Alignment creates order and connection. Left alignment is easiest to read on screens.
Left Aligned
This text is aligned to the left. It creates a clean reading flow.
Center Aligned
This text is centered. Best for headings and short text.
Right Aligned
This text is aligned to the right. Use it sparingly.
Justified
This text is justified. It aligns both left and right margins for a clean block of text.
4. READABILITY
Choose readable fonts, use good line length and spacing, and ensure clear letterforms.
GOOD
Easy to read with proper spacing and line length.
The quick brown fox jumps over the lazy dog.
POOR
Hard to read due to tight spacing and long lines.
The quick brown fox jumps over the lazy dog. The quick brown fox jumps over the lazy dog. The quick brown fox jumps over the lazy dog.
5. FONT PAIRING
Pair fonts that complement each other. Combine contrast with harmony.
GOOD PAIRINGS
Playfair Display
Montserrat
Raleway
Merriweather
Lora
Open Sans
Poppins
Lato
POOR PAIRINGS
Comic Sans
Papyrus
Impact
Curlz MT
6. WHITESPACE
Whitespace (or negative space) gives content room to breathe and improves focus.
Clean & Clear
Good use of whitespace makes content feel organized and easy to read.
Crowded & Busy
Too little whitespace makes content feel cluttered and overwhelming.
7. CONSISTENCY
Be consistent with fonts, sizes, colors, and styles across all designs.
Font Choices
Sizes & Weights
Colors
Styles & Formatting
Consistency builds a strong visual identity and improves user experience.
8. LIMIT FONT STYLES
Use a maximum of 2–3 fonts and limit weights and styles to keep designs clean and professional.
DO
Aa Aa Aa
2 Fonts
(with a few weights)
Clean. Focused. Professional.
DON'T
Aa Aa Aa
Aa Aa Aa Aa
Too many fonts create confusion.
9. COLOR & TYPOGRAPHY
Ensure enough contrast between text and background for accessibility.
GOOD CONTRAST
Aa
Easy to read
POOR CONTRAST
Aa
Hard to read
Aim for high contrast to make content accessible to everyone.
10. PURPOSE & TONE
Choose fonts that match the purpose and tone of your message.
Professional
Montserrat / Lato
Friendly
Roboto / Nunito
Creative
Playfair Display / Poppins
Elegant
Cormorant / Lora
Great typography is invisible. You don't notice the type—you notice the message.

Ps
DESIGN BASICS
BETTER FONTS. BETTER DESIGNS.
10 Font Combinations Every Designer Should Know
PERFECT PAIRINGS / BETTER HIERARCHY / STRONGER DESIGNS
Great design isn't just about what you say, but how you say it.
1
Modern Living
SIMPLE SPACES. BETTER DAYS.
Montserrat + Lora
A clean and elegant combination. Montserrat brings the modern, bold presence, while Lora adds a touch of sophistication.
Montserrat (Bold) + Lora (Regular)
2
The Creative Mindset
IDEAS / DESIGN / IMPACT
Playfair Display + Montserrat
A classic and modern duo. Playfair Display gives a stylish, high-end feel, while Montserrat keeps it clean and readable.
Playfair Display (Bold) + Montserrat (Regular)
3
GOOD THINGS Take Time
STAY PATIENT / STAY GROWING
Bebas Neue + Poppins
Bebas Neue adds strength and impact, while Poppins balances it with a clean and friendly look.
Bebas Neue (Bold) + Poppins (Regular)
4
Minimal & Clean
LESS CLUTTER. MORE IMPACT.
Cormorant Garamond + Helvetica
Cormorant Garamond brings elegance and style, while Helvetica adds clarity and balance for a clean finish.
Cormorant Garamond (Medium) + Helvetica (Regular)
5
Nature Heals
BREATHE / EXPLORE / RECONNECT
Fraunces + Inter
Fraunces gives a bold, organic feel, while Inter keeps it modern and easy to read. A great mix of character and clarity.
Fraunces (Bold) + Inter (Regular)
6
BIG DREAMS
SMALL STEPS
Anton + Open Sans
Anton creates a strong visual impact, while Open Sans keeps the message clean and easy to read.
Anton (Bold) + Open Sans (Regular)
7
Better Together
IDEAS / PEOPLE / PROGRESS
Cormorant Infant + Roboto
Cormorant Infant adds a stylish, creative touch, while Roboto ensures good readability and a clean layout.
Cormorant Infant (Bold) + Roboto (Regular)
8
URBAN STYLE
BOLD LOOKS. REAL PEOPLE.
Bebas Neue + Raleway
Bebas Neue brings the bold, urban energy, while Raleway adds a clean, minimal and modern balance.
Bebas Neue (Bold) + Raleway (Regular)
9
FRESH IDEAS
FOR A BRIGHTER TOMORROW
Archivo Black + Lora
Archivo Black gives a strong, confident headline, while Lora adds a refined and human touch.
Archivo Black (Bold) + Lora (Regular)
10
Higher Goals
DREAM / PLAN / ACHIEVE
Space Grotesk + Merriweather
Space Grotesk feels modern and techy, while Merriweather adds readability and a timeless, classic vibe.
Space Grotesk (Bold) + Merriweather (Regular)
QUICK TIPS
Use a bold font for headlines.
Choose a simpler font for body text.
Maintain contrast between the fonts.
Keep it consistent across your designs.
GOOD FONTS / GOOD VIBES / GREAT

# COLOR IN WEB DESIGN

- Large websites often use a different color for each section, making it easier for users to identify where they are.
- A consistent color scheme helps improve navigation and usability.
- Be consistent and use the same color for the same purpose throughout the site.
- Aim to create a sense of visual harmony.
- Consider the principles of color psychology when selecting colors.

# COLOR IN WEB DESIGN (II)

| Color | Associated Meaning |
| --- | --- |
| Blue | Trustworthy Business |
| Red | Love |
| Orange | Fun |
| Pink | Fashion |
| Green | Nature |

# ICONOGRAPHY IN WEB DESIGN

- Icons represent the actions or functions they are associated with.
- Maintain a consistent visual style across all icons, pages, and sections of the website.
- Icons help reduce the need for excessive text, making interfaces cleaner and easier to use.
- Choosing the right icons is essential for effective communication.
- If users cannot easily understand an icon's meaning, it fails to save them time and may create confusion.

# HTML MARKUP LANGUAGE REVIEW

- HTML (HyperText Markup Language) is the standard markup language used to create web pages.
- It presents content in a structured and user-friendly format.
- HTML supports:
  - Hyperlinks to connect documents and web pages.
  - Multimedia content, such as images, audio, and video.

- The core purpose of HTML is to define the logical structure of content, including:
  - Headings
  - Paragraphs
  - Lists
  - Definitions
  - Quotations

# KEY FEATURES OF HTML

a) Simple and easy to learn.

b) Does not use variables like traditional programming languages.

c) Not compiled.

d) Interpreted directly by the web browser.

e) Instructions are written as tags (elements).

f) Enables the creation of hypertext through hyperlinks.

# BASIC STRUCTURE OF AN HTML DOCUMENT

- An HTML document is defined by **tags**, which are reserved words used to structure and format content.
- The entire document is enclosed within the <html> tag.
- HTML tags divide the document into different sections and define their purpose.
- An HTML page consists of two main parts:
  - **Head** (<head>): Contains metadata, title, links to stylesheets, and other configuration information.
  - **Body** (<body>): Contains the visible content displayed in the browser.

```
<html>
  <head>
    …
  </head>
  <body>
    …
  </body>
</html>
```

# BASIC HTML ELEMENTS: TEXT, PARAGRAPHS, STYLES, AND FORMATTING

- Text displayed on a web page is placed between HTML tags, which define its style and formatting.

- Tags help organize content and control how it appears in the browser.

- One of the most commonly used tags is the paragraph tag:
  -  opens a paragraph.
  -  closes a paragraph.

```
<p><b>Hola, estamos escribiendo en un párrafo.</b></p>
<p>Ahora ya hemos cambiado de párrafo.</p>
```

- Paragraphs automatically:
  - Start on a new line.
  - Add spacing before the next block of content.

**Hola, estamos escribiendo en un primer párrafo.**
Ahora ya hemos cambiado de párrafo.

# TAGS IN HTML

- **...** Displays text in bold (in the past it was ...)
- **...** Displays text in italic (in the past it was ... )
- **<u>...</u>** Underlines text
- **<sub>...</sub>** Displays text as subscript
- **<h1>...</h1>** Main heading (highest level)
- **<h2>...</h2>** Second-level heading
- **<h3>...</h3>** Third-level heading
- **<h4>...</h4>** Fourth-level heading

# LINKS

- The **<a>** (anchor) tag is used to create hyperlinks in HTML.
- It can be used in two main ways:

1. **Linking to a resource or web page**
    - Uses the href attribute followed by the URL.
    - Allows users to navigate to another page, file, or website.

2. **Creating a destination within a page**
    - Traditionally used the name attribute to identify a section that other links could reference.
    - In modern HTML, the **id** attribute is generally preferred.

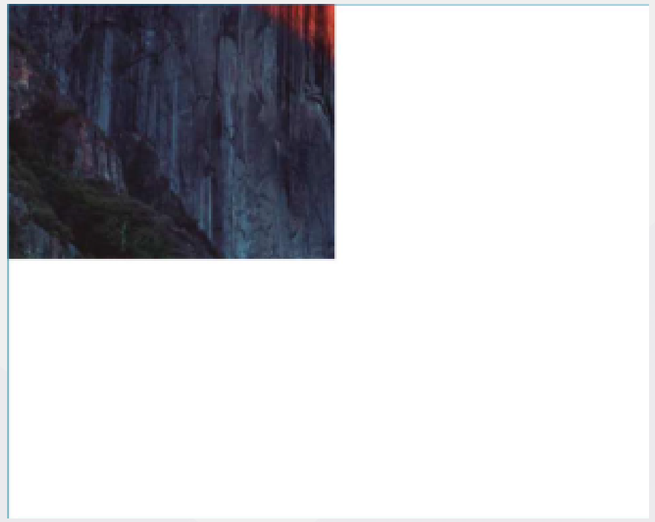
```
<body>
  <p><a href="https://www.sintesis.com">Editorial Síntesis</a></p>
</body>
```

# IMAGES

- Images make a website more attractive and engaging than text-only content.
- High-quality images should be used to enhance the visual experience.
- Images can help communicate information more effectively and improve user engagement.
- When using images from external sources, it is important to respect copyright and licensing requirements.
- Always ensure that images are either:
  - Created specifically for the website, or
  - Used with the appropriate permission or license.

# IMAGES (II)

```
<html>
 <head>
  <title>EJERCICIO 2.3. EJEMPLO CON IMÁGENES (CAPÍTULO 3)</title>
 </head>
 <body>
  <img src="ejemplo.png" alt="Imagen fondo pantalla">
 </body>
</html>
```



# FORMS

- Forms are interactive elements used to collect information from users on a website.
- The data entered is sent to a server, where it is processed and stored.
- A form can contain different types of input controls to gather information in various ways.
- Forms enable users to interact with websites and submit data efficiently.
- Example Uses:
  - Contact forms
  - Registration forms
  - Surveys and questionnaires
  - File uploads

# THE <INPUT> ELEMENT

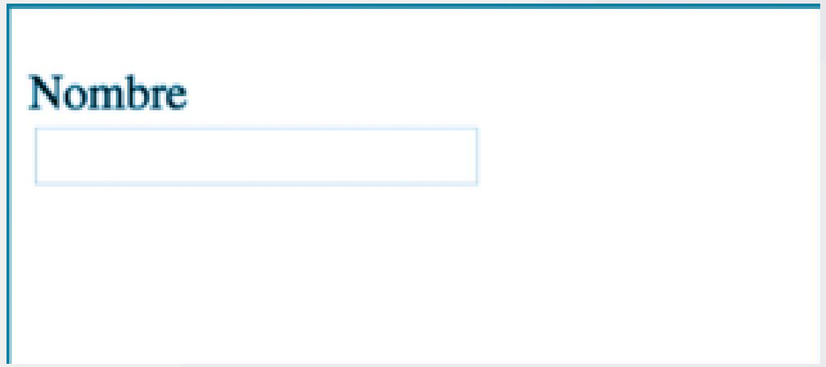
- The <input> element is one of the most important and widely used form controls in HTML.
- It is used to create interactive fields that allow users to enter or select data.
- The behavior of an <input> element is determined by its type attribute.
- By default, the value of the type attribute is text, which creates a single-line text field.
- Different type values provide different kinds of input controls.

# COMMON INPUT TYPES

- **text** – Single-line text input
- **password** – Password field with hidden characters
- **email** – Email address input
- **number** – Numeric input
- **radio** – Radio button
- **checkbox** – Checkbox
- **file** – File upload control
- **submit** – Form submission button

# TEXT

```
<html>
 <head>
  <title>…</title>
 </head>
 <body>
      <form action="" method="get">
       <br>Nombre<br/>
      <input type="text" name="nombre" value=""/>
      </form>
 </body>
</html>
```



# BUTTON

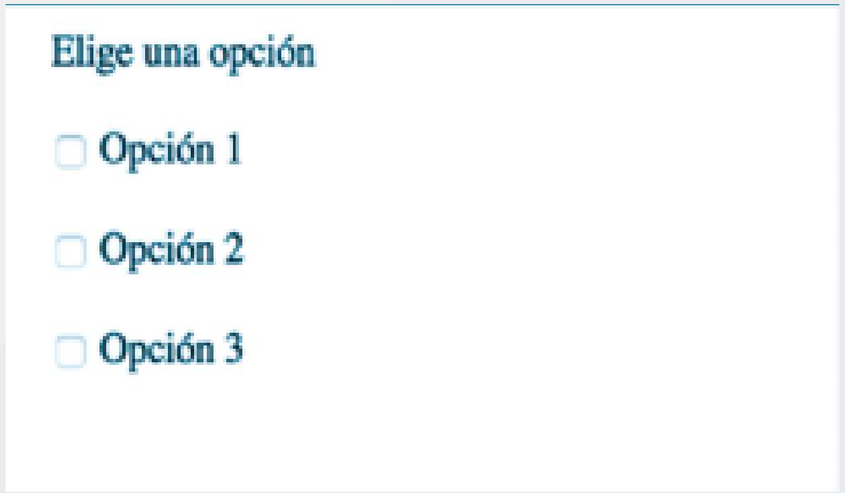
A distinction must be made between the submit action performed by buttons created using the "**submit**" type and other buttons intended for different operations. These other buttons must be implemented using JavaScript and can be programmed to perform any task, no matter how complex.

```
<html>
 <head>
  <title>…</title>
 </head>
 <body>
      <form action="" method="get">
       <input type="button" name="guardar" value="Guardar"/>
      </form>
 </body>
</html>
```

# CHECKBOX

A selectable option that uses the "value" attribute to specify the value submitted with the form. The "checked" attribute determines whether the checkbox is selected by default.

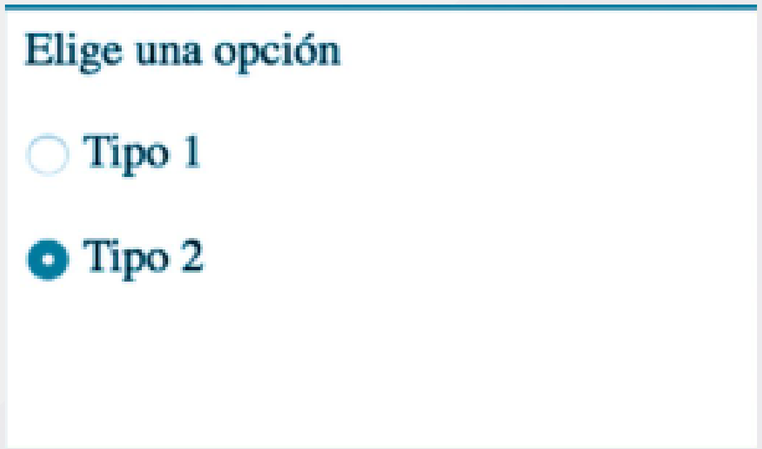
```
<html>
 <head>
 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8"/>
  <title>…</title>
 </head>
 <body>
      <form action="" method="get">
        <br>Elige una o varias opciones</br>
        <br><input name="op1" type="checkbox" value="op1"/>Opción 1
        </br>
        <br><input name="op2" type="checkbox" value="op2"/>Opción 2
        </br>
        <br><input name="op3" type="checkbox" value="op3"/>Opción 3
        </br>
      </form>
 </body>
</html>
```



# RADIO

Used when only one option can be selected from a group. Selecting one option automatically deselects the others. Radio buttons are grouped using the "name" attribute.

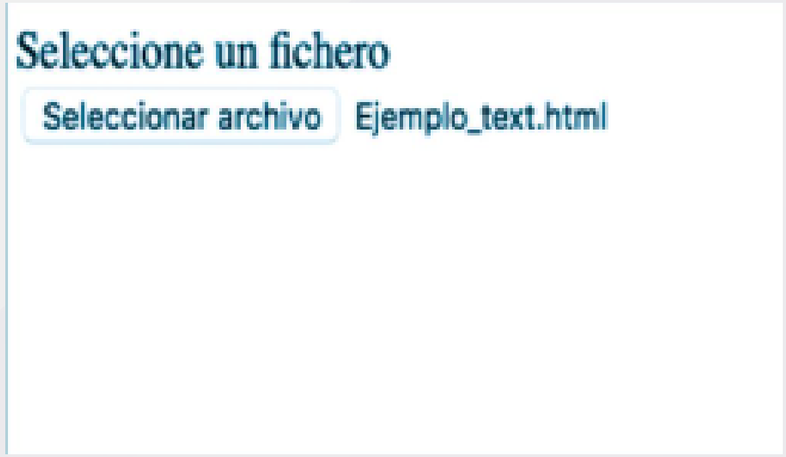
```
<html>
 <head>
 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8"/>
  <title>…</title>
 </head>
 <body>
      <form action="" method="get">
        <br>Elige una opción<br/>
        <br><input  type="radio"  name="Tipo"  value="tipo1"/>Tipo  1
        </br>
        <br><input  type="radio"  name="Tipo"  value="tipo2"/>Tipo  2
        </br>
      </form>
 </body>
</html>
```



# FILE

- Allows the user to select a file from their device.
- The "accept" attribute can be used to specify the types of files that can be selected.
- When the user clicks the button, a file browser window opens.
- The user can then navigate the system and choose the desired file.

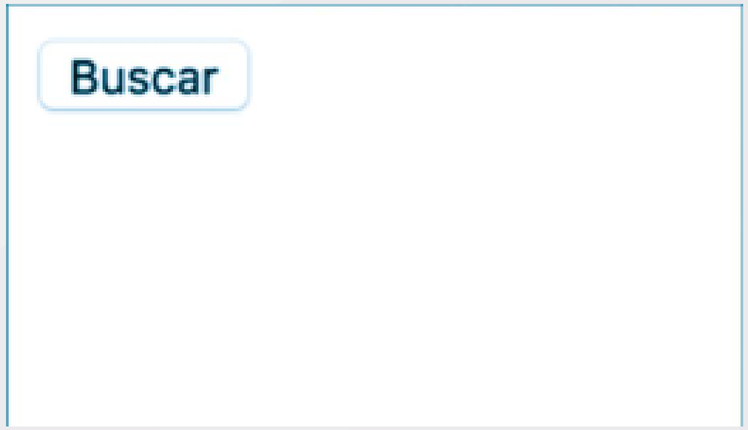
```
<html>
 <head>
  <title>…</title>
 </head>
 <body>
      <form action="" method="get">
        Seleccione un fichero
        <br><input type="file" name="adjunto"/></br>
      </form>
 </body>
</html>
```



# SUBMIT

- Sends the form data entered by the user to the server.
- Can display custom text defined by the developer.
- If no text is specified, the browser displays a default label.
- Commonly used to submit forms for processing.

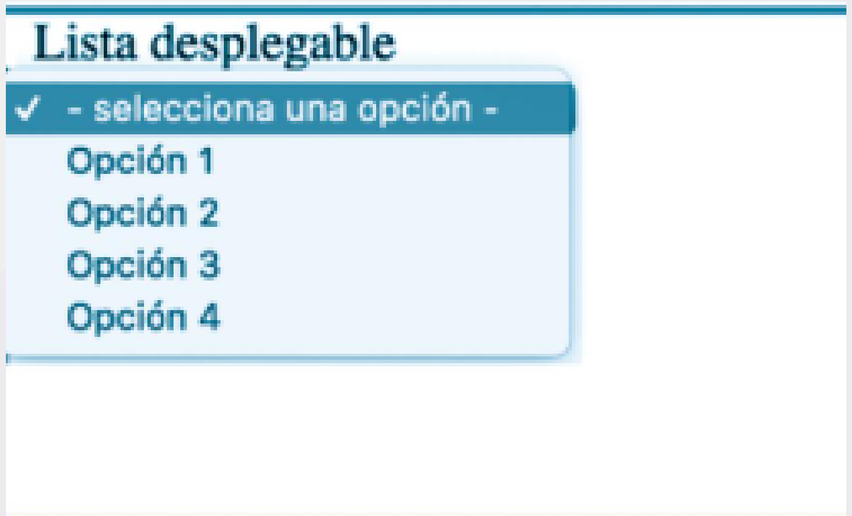
```
<html>
 <head>
  <title>…</title>
 </head>
 <body>
      <form action="" method="get">
        <input type="submit" name="buscar" value="Buscar"/>
      </form>
 </body>
</html>
```



# SELECT

- Used to create **drop-down lists** in a form.
- Contains one or more options defined with the **<option>** tag.
- Allows users to select an item from a predefined list.
- The number of available options can be customized as needed.

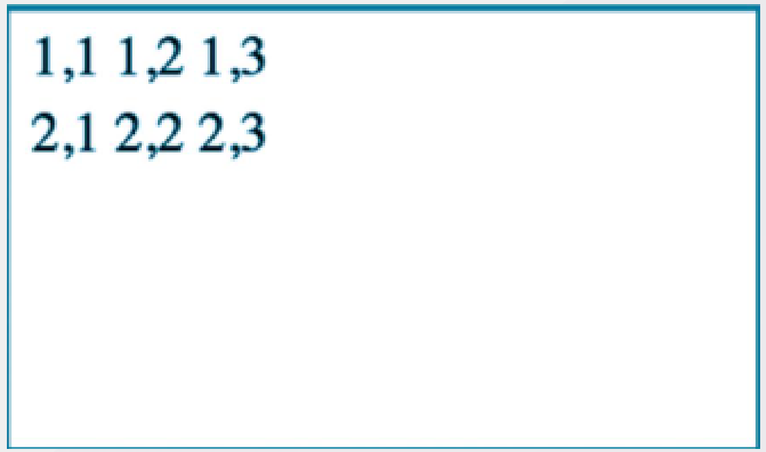
```
<html>
 <head>
  <meta http-equiv="Content-Type" content="text/html; charset=UTF-8"/>
  <title>…</title>
 </head>
 <body>
  <label for="lista">Lista desplegable</label><br/>
  <select id="lista" name="lista">
     <option value="" selected="selected">- selecciona una opción -</
     option>
     <option value="opcion1">Opción 1</option>
     <option value="opcion2">Opción 2</option>
     <option value="opcion3">Opción 3</option>
     <option value="opcion4">Opción 4</option>
  </select>
 </body>
</html>
```



# TABLES

- Tables are used to display information in a clear and organized way.
- To create a table, you first define its structure and properties.
- The table is then built row by row.
- Each row contains one or more cells, which store the data for each column.
- Tables are useful for presenting structured data such as schedules, prices, statistics, and comparisons.

```
<table>
  <tr>
    <td>1,1</td>
    <td>1,2</td>
    <td>1,3</td>
  </tr>
  <tr>
    <td>2,1</td>
    <td>2,2</td>
    <td>2,3</td>
  </tr>
</table>
```



# MAIN TABLE ATTRIBUTES IN HTML

| Attribute | Purpose |
| --- | --- |
| `border` | Defines the thickness of the table and cell borders. The default value is `0`, which means no border. |
| `width` | Specifies the width of the table. It can be set as a percentage of the available page width (e.g., `100%`) or in pixels. |
| `align` | Defines the table alignment: `left`, `right`, or `center`. |
| `caption` | Adds a descriptive title displayed above the table. |

# TABLE CAPTIONS

- Tables often include a title to help users quickly identify the content being displayed.
- To add a title to a table, use the **<caption>** tag.
- The caption text appears at the top of the table and provides a clear description of its contents.

```
<table border=1 width="50%" align=center>
 <caption>
      Título de la tabla
 </caption>
 <tr>
      <td>1,1</td>
      <td>1,2</td>
      <td>1,3</td>
 </tr>
 <tr>
      <td>2,1</td>
      <td>2,2</td>
      <td>2,3</td>
 </tr>
</table>
```

# FRAMES

- Although **HTML5 has made frames obsolete**, it is useful to understand how they work because some legacy websites still use them and may require maintenance or improvements.

- Frames are **independent windows embedded within a web page**, dividing the page into multiple subpages.

- This approach helps create a more organized and visually clean layout, as different sections can display separate content simultaneously.

- Unlike a standard HTML document, which contains <head> and <body> sections, a frame-based document uses a <frameset> element instead of the <body> element.

# FRAMES (II)

- To create frames, the <frameset> tag is used.
- The <frameset> element defines the layout of the page by specifying how many frames will be displayed and how they are arranged.
- Typically, the number and size of the columns or rows within the main page are defined first.
- The <frame> tag is then used to configure each frame individually, specifying the content and properties of each section.

# FRAMES (III)

```
<frameset cols=anchoColumna1, anchoColumna2, …anchoColumnaN>
 <frame …>
 <frame …>
</frameset>
```

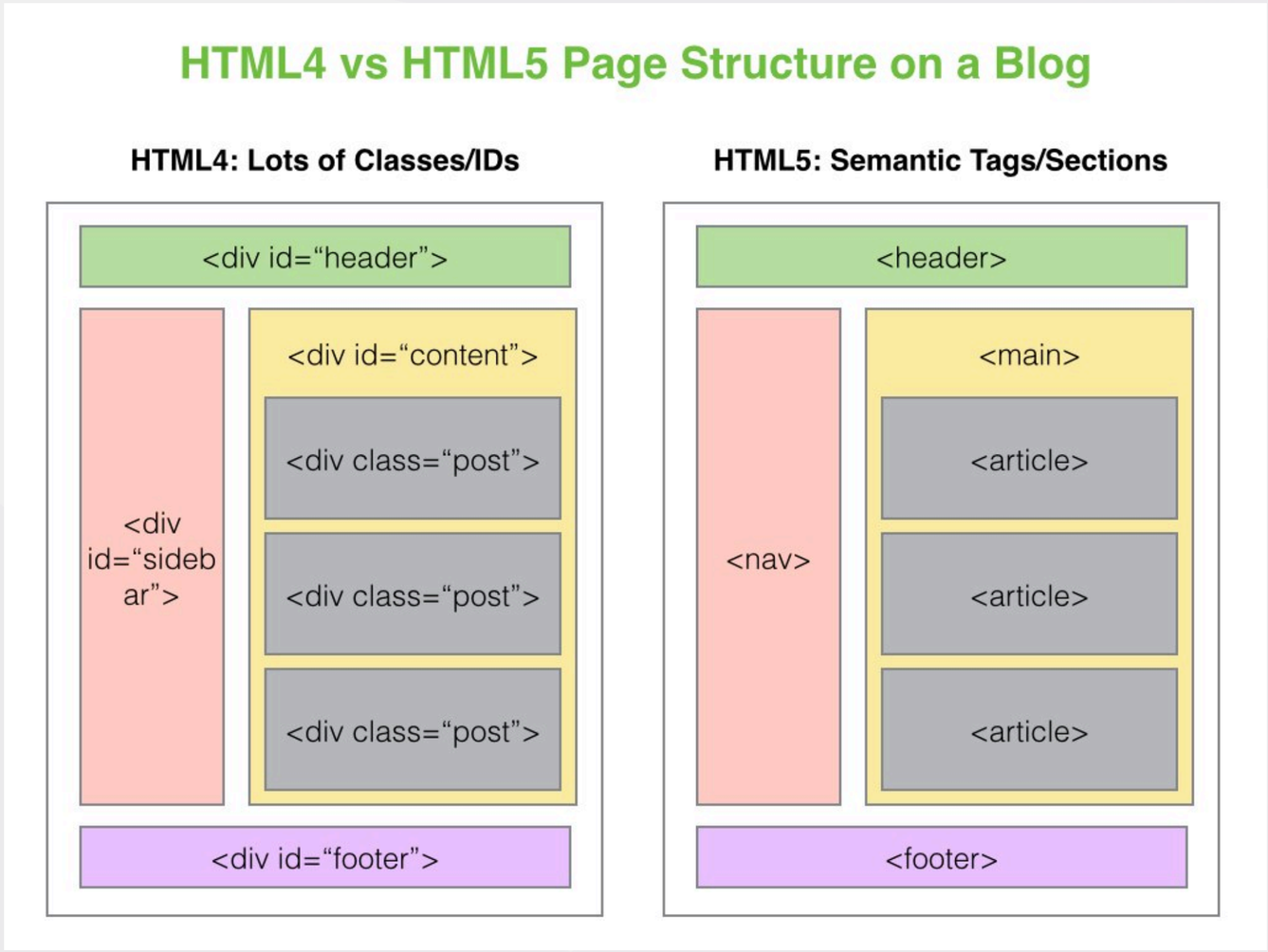
```
<frameset cols=10%,20%,70%>
 <frame src="1">
 <frame src="2">
 <frame src="3">
</frameset>
```

```
<frameset cols=30%,70%>
 <frame src="1">
 <frameset rows=40%,*>
  <frame src="2">
  <frame src="3">
 </frameset>
</frameset>
```

# HTML5

- HTML5 is the latest version of the HTML language used for web development.
- Websites built with HTML5 are best viewed in modern browsers, as older browsers may not support some of its newer elements.
- One of HTML5's key features is the introduction of **semantic elements** (or semantic markup).
- Semantic elements divide a document into meaningful sections and clearly describe their purpose to the browser.
- Unlike the generic  element, semantic tags provide information about the role of the content they contain to the browser and enhance accessibility for users with assistive technologies.

# HTML5 (II)


HTML4 vs HTML5 Page Structure on a Blog
HTML4: Lots of Classes/IDs

HTML5: Semantic Tags/Sections
<header>
<nav>
<main>
<article>
<article>
<article>
<footer>

# NEW SEMANTIC ELEMENTS IN HTML5: <ARTICLE>

- The <article> element is used to represent an independent unit of content within a web page.

- A page can contain multiple <article> elements, depending on the amount of content being presented.

- Each article can be divided into several sections to improve structure and organization:
  - <header>: Article heading or introductory information.
  - <section>: Main content of the article.
  - <footer>: Closing information, such as author details, references, or publication dates.

# NEW SEMANTIC ELEMENTS IN HTML5: <ASIDE>

- The <aside> element is used to define the sidebar of a web page.

- Although it can contain any type of content, it is commonly used for supplementary information related to the main content.

- Typical uses include:
  - Links to other websites
  - Social media links
  - Related articles
  - Advertisements
  - Additional resources or references

```
<aside>
 <h3>Barra lateral de navegación</h3>
 <a href="http:www.google.com"><b> Buscador de Google </b></a>
</aside>
```

# NEW SEMANTIC ELEMENTS IN HTML5: <FOOTER>

- The <footer> element defines the footer section of a web page.

- It is commonly used to display:
  - Copyright information
  - Contact details
  - Website credits
  - Contact forms

- Legal notices and related information.

- A <footer> can be used for the entire website or for specific content sections, such as an <article>.

# NEW SEMANTIC ELEMENTS IN HTML5: <HEADER>

- The <header> element defines the header section of a web page.
- It represents the top part of the page and usually contains branding elements that identify the website.
- In most cases, the header includes the navigation menu (<nav>).
- It may also contain logos, titles, search bars, or introductory content.

# NEW SEMANTIC ELEMENTS IN HTML5: <MAIN>

- The <main> element represents the primary content of an HTML document's <body>.
- This content is directly related to the main topic or purpose of the page.
- The content inside <main> should be unique to that document and should not include content repeated across multiple pages.
- An HTML document should contain only one <main> element.
- The <main> element can include other semantic elements such as <article>, <aside>, <section>, <header>, <footer>, and <nav>.

```
<main>
  <nav>…</nav>
  <p>…</p>
</main>
```

# NEW SEMANTIC ELEMENTS IN HTML5: <NAV>

- The <nav> element creates a navigation menu that links to different sections of a website.
- It typically contains the main navigation links used to move between pages or sections.
- The navigation menu is usually placed:
  - Directly below the header (<header>), or
  - On the left side of the page.

# NEW SEMANTIC ELEMENTS IN HTML5: <SECTION>

- The <section> element is used to create distinct sections within a web page.
- It can be considered a semantic alternative to the generic  element.
- Each section groups related content and helps organize the structure of the page.
- A <section> may contain one or more <article> elements.

# NEW SEMANTIC ELEMENTS IN HTML5: <TIME>

- The <time> element is used to represent a date, a time, or both within a web page.

- It helps browsers and search engines recognize and process temporal information more accurately.

# NEW FORM ELEMENT IN HTML5: TYPE="EMAIL"

- HTML5 introduces the email input type, which creates a field specifically for entering an email address.
- The browser automatically validates the format of the email address entered by the user.
- This reduces the need for additional JavaScript code to perform basic email validation.

# NEW SEMANTIC ELEMENTS IN HTML5: <MARK>

- The <mark> element is used to highlight text within a document.
- Highlighted text is visually emphasized to indicate its importance or relevance in a particular context.
- By default, browsers usually display marked text with a highlighted background color

```
<main>
  <h2>Bienvenido a <mark>HTML5</mark></h2>
</main>
```

HTML5 introduced the mark element for highlighting text.

# NEW SEMANTIC ELEMENTS IN HTML5: <FIGURE>

- The <figure> element represents self-contained visual content within a document.

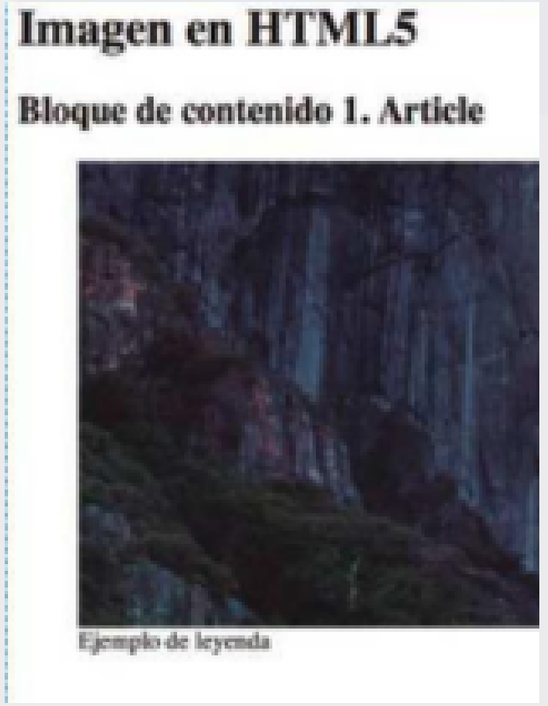
- It is commonly used for:
  - Images
  - Illustrations
  - Diagrams
  - Code snippets
  - Charts and other visual content

- A <figure> can be moved to another part of the document, or even to an appendix, without affecting the main flow of the content.

# NEW SEMANTIC ELEMENTS IN HTML5: <FIGCAPTION>

- The <figcaption> element provides a caption or title for a figure.
- It is placed inside the <figure> element and is associated with the content of that figure.
- The caption helps users understand the meaning or context of the visual content.

```
<html>
 <head>
  <title>Título de la WEB</title>
  <meta charset="UTF-8">
 </head>
 <body>
  <header>
   <h1><b>Imagen en HTML5</b></h1>
  </header>
  <section>
   <article>
    <h2>Bloque de contenido 1. Article</h2>
    <figure>
     <img src="ejemplo.png" alt="ejemplo">
     <figcaption>Ejemplo de leyenda</figcaption>
    </figure>
   </article>
  </section>
 </body>
</html>
```



# NEW SEMANTIC ELEMENTS IN HTML5: <DETAILS>

- The <details> element is used to display additional information that can be shown or hidden by the user.
- It creates an interactive widget that expands and collapses content, similar to an accordion menu.
- By default, the extra information remains hidden until the user opens the element.

# NEW SEMANTIC ELEMENTS IN HTML5: <SUMMARY>

- The <summary> element provides a heading or summary for a <details> element.
- It acts as the clickable title that users select to show or hide the additional content.
- Only one <summary> element should be associated with a <details> element.

# <DETAILS> & <SUMMARY>

```
<html>
  <head>
        <title>Título de la WEB</title>
        <meta charset="UTF-8">
  </head>
  <body>

        <details open>
              <summary>Prueba de summary en HTML5</summary>
              <p>TEXTO QUE SE MUESTRA AL INCORPORAR UN
               ELEMENTO DETAILS EN HTML5 Y DESPLEGARLO</p>
        </details>
  </body>
</html>
```

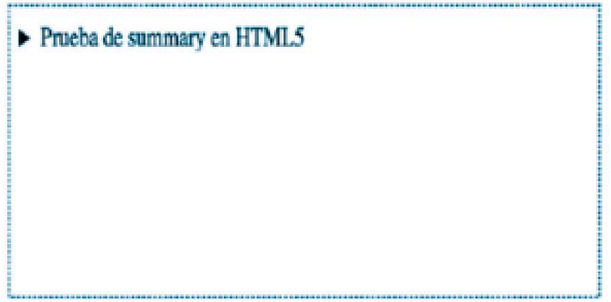



**Figura 3.15**
Antes de desplegar el menú.

**Figura 3.16**
Después de desplegar el menú.

# WEB USABILITY, ACCESSIBILITY AND NAVIGATION: ANALYSIS AND VERIFICATION

**Objectives:**

- Recognize the need to design accessible websites and **identify the main accessibility guidelines** for content.
- Analyze the accessibility of different websites and web documents.
- Assess the usability of different websites and web documents.
- Evaluate the importance of using **standards** in web page development.
- Modify and design web documents to adapt them to the objectives and intended users.
- Verify the ease of navigation of a website.

# WEB USABILITY

In the digital age and the era of ICT (***I****nformation and* ***C****ommunications* ***T****echnology*), the amount of content provided through the web has increased exponentially.

Furthermore, users now have greater familiarity with the use of new technologies, including the web.

As a result, modern websites feature an appearance and functionality that would have been difficult to imagine just a few years ago.

# WEB USABILITY (II)

"*If you don't make it easy, users will leave your website.*"- Jakob Nielsen

For a website to be successful, it is becoming increasingly important to provide **quick access to content** and deliver a **pleasant visual experience to users**, encouraging them to return in the future to obtain the information they need.

The success or failure of a website largely depends on **user satisfaction**.

This satisfaction is influenced by a number of factors related to the clarity and usefulness of the content, ease of access, overall design quality, and more.

An appropriate web design is essential to facilitate the exchange of information with users.

# OBJECTIVES OF USE AND USABILITY STANDARDS

- Usability is a concept intrinsically related to the way a web page is presented to users, as well as to the way users interact with it.

- It takes into account factors such as simplicity, clarity, and so on.

- **ISO (International Organization for Standardization)** defines usability as the ability of a particular software product to be understood, used, and learned by users, while also being attractive to them.

- The father of this concept is considered to be **Jakob Nielsen**, author of the previous quote. Nielsen defined usability as a concept related to the quality of a product, measuring how easy it is to use.

# OBJECTIVES OF USE AND USABILITY STANDARDS (II)

- A website must not only have an attractive appearance and be technologically advanced, but it must also be easily understandable and comprehensible to any user, in order to provide the information they are looking for in the shortest possible time.

- Usability depends on the product itself, but also on the user, specifically on how they interact with the website and how they perceive its simplicity and ease of use.

- There are **subjective parameters** (user satisfaction) and **objective parameters** (time taken by the user to achieve a goal, errors made while trying to achieve it, etc.) that can be used to measure the usability of a particular website.

# OBJECTIVES OF USE AND USABILITY STANDARDS (III)

**How can a website achieve good usability?**

To achieve this, before starting a project, it is important to consider questions such as the following:

- What is being offered to the user?
- Who are the potential users, and what level of education or knowledge will they have?
- What information will users be looking for?
- In what context will potential users interact with the website?

# ISO 9241: ERGONOMICS OF HUMAN-SYSTEM INTERACTION

- International standard that addresses the ergonomics of human-computer interaction (HCI).

- It provides guidelines and principles for the design and evaluation of interactive systems, taking usability and user experience into account.

- It is organized into various **parts of the standard**:

  - **ISO 9241-11:** *Principles of usability and user-centered design*. Provides general principles and guidelines for user-centered design.
  - **ISO 9241-210:** *Human-centred design process for interactive systems.* Describes an iterative, user-centered design process for interactive systems.
  -

# ISO 9241: ERGONOMICS OF HUMAN-SYSTEM INTERACTION (II)

- Parts of the standard (II):

  - **ISO 9241-110:** *Dialogue principles*. Focuses on the design of dialogues between users and systems. The user-system dialogue principle is based on smooth and effective communication between the user and the system during interaction. Recommended considerations include immediate feedback, clarity in information presentation, and consistency in the interface, among others.
  - **ISO 9241-12:** *Presentation of information*. Covers the visual presentation and organization of information in user interfaces.
  - **ISO 9241-171:** *Accessibility guidance for software.* Provides specific guidelines to improve accessibility in human-computer interaction.

# ISO/IEC 25000

- The ISO/IEC 25000 family of standards, known as **SQuaRE** (*System and Software Quality Requirements and Evaluation*), is a set of standards aimed at creating a common framework for evaluating the quality of software products.

- The ISO/IEC 25000 family is the result of the evolution of previous standards and is composed of five divisions

# ISO/IEC 25010

- This section focuses on the **software quality model**.
- This model serves as the cornerstone for establishing a system for evaluating product quality. It defines the quality characteristics that should be considered when assessing the properties of a software product.
- Software product quality can be interpreted as the degree to which a product meets the requirements of its users.
- The product quality model defined by ISO/IEC 25010 is composed of **eight quality characteristics:** functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability

# ISO/IEC 25010 (II)

# USABILITY VS ACCESSIBILITY

- Specifically, usability is defined as the ability of a software product to be understood, learned, and used, as well as to be attractive to the user when employed under certain conditions. Some of the sub-characteristics directly related to usability include learnability, operability, protection against user errors, and the aesthetics of the user interface.

- While **usability** focuses on the overall user experience and effectiveness of interaction, **accessibility** focuses on ensuring that information and functionality are available to everyone, including people with diverse abilities and limitations. Both concepts are essential for creating products and services that are effective, efficient, and accessible to a broad audience.

# OTHER STANDARDS AND REGULATIONS

Their purpose is to achieve consistency in design criteria, since it would not be logical for each web designer to choose different usability parameters:

- **ISO/IEC 9126**. An international standard for software quality evaluation. It is divided into four parts: quality model, external metrics, internal metrics, and quality-in-use metrics

- **ISO 13407**. Provides a guide for achieving quality in use through the incorporation of iterative activities involved in **User-Centred Design** (UCD).

- **ISO 9241/151**. *Ergonomics of human-system interaction. Part 151*: Guidance on World Wide Web user interfaces. It provides recommendations on user-centred design for web user interfaces in order to improve usability.

# OTHER STANDARDS AND REGULATIONS (II)

- **UNE 139803:2004**. Titled *Computer Applications for People with Disabilities. Accessibility Requirements for Web Content*. This Spanish standard, published in December 2004, contains the specifications that web content must meet to be considered accessible. It is based on the **Web Content Accessibility Guidelines** (WCAG 1.0) developed by the W3C Initiative, although structured differently.

- **UNE 139803:2012.** Due to the differences between WCAG 2.0 and WCAG 1.0, it became necessary to update the content of the UNE standard so that its requirements aligned with WCAG 2.0. As a result, in 2012 the standard was revised to directly adopt WCAG 2.0. This updated UNE standard specifies which parts of WCAG 2.0 are considered mandatory requirements and their corresponding priority levels.

# USERS

Usability must take into account the wide variety of abilities, as well as the great cultural, cognitive, and psychological diversity that exists.

Some of the most common differences are:

- **Cognitive and perceptual abilities:** Language comprehension, learning capacity, concept assimilation, problem-solving skills, etc.
- **Cultural factors:** Linguistic diversity or cultural background. These may affect the interpretation of formats, units of measurement, signs, punctuation marks, protocols, and formalities.
- **Disabilities:** One of the most important aspects to consider during the website development process regarding usability and, in particular, accessibility, is adapting the developed application to people with disabilities.
- **Technological factors:** Internet connection, screen sizes, memory requirements, and processing capabilities.

# TYPES OF WEBSITE USERS

According to Their Access Resources and Purpose

An **Anonymous User** is someone who navigates the site without identifying themselves as a registered user or creating an account. For example, someone who has not access to the private area.

A **Registered End User** is a user who has signed up on a website. This is generally intended to provide certain privileges or store session data in order to streamline the browsing process. For example, access to the private area of a banking website.

**Beta Tester User:** During the software development process, a user profile is often created for a beta tester, whose role is to perform tests and verify that the website functions according to the client's requirements. They report any issues they find to the website designers, who are responsible for resolving them before the final deployment.

# WEB NAVIGATION

The foundation of good web interface design is user interaction with the site.

The following concepts must be taken into account:

a) **Accessible information:** Users can access it easily and conveniently. To achieve this, content must be properly organized, since if the website’s navigation system is not simple, users may struggle to find information and are likely to leave the site.

b) **Consistency:** users can understand and use its features without needing instructions. Consistency is related to the coherence among the elements that make up the site's structure.

# WEB NAVIGATION (II)

Concepts (II):

c) **Persistence.** It refers to the arrangement of the different elements that make up a website. The appearance and location of page components (header, navigation bar, content area, etc.) should remain similar across all pages within the site.

d) **Simplicity of navigation.** This concept is closely linked to website navigability, as simple navigation allows users to know at all times where they are within the site and where they can go next. To achieve this, the website should be consistent and persistent, while also using clear language for links and menus.

# ACCESSIBILITY

The concept of accessibility is directly related to usability.

**Accessibility** can be defined as the ability to access a particular website, whereas usability refers to its ease of use. Therefore, it is clear that a website must be accessible before it can be usable.

This essentially means that access to a website should be facilitated for all users, regardless of the technical limitations of their devices (software, hardware, etc.) or their individual limitations (disabilities, proficiency in a particular language, etc.).

Consequently, an accessible website must take into account the wide diversity of potential users it may have.

# THE WORLD WIDE WEB CONSORTIUM (W3C)

The World Wide Web Consortium (W3C) is an international community in which member organizations **develop standards that ensure the growth and accessibility of the Web**. It was created in 1994 with the aim of developing interoperable technologies.

Within this framework, it became necessary to develop strategies, guidelines, and resources to guarantee equal access to the Web. This led to the creation of the Web Accessibility Initiative (WAI). The initiative developed the Web Content Accessibility Guidelines (WCAG) 2.0, which provide recommendations and techniques for making software and web content accessible. These guidelines were approved as the international standard ISO/IEC 40500:2012. In June 2018, WCAG 2.0 was superseded by WCAG 2.1

# THE WORLD WIDE WEB CONSORTIUM (W3C) (II)

On 5 October 2023, a significant milestone in web accessibility was reached with the publication of the updated **Web Content Accessibility Guidelines (WCAG) 2.2**, whose latest revision further raises the standards for creating inclusive digital content.

WCAG 2.2 aims to:

- Improve accessibility for users with cognitive and learning disabilities.
- Enhance the experience for users with low vision.
- Optimize navigation on mobile devices.

The latest version introduces nine new success criteria.

# THE WORLD WIDE WEB CONSORTIUM (W3C) (II)

The four principles that govern this framework are that web design must be perceivable, operable, understandable, and robust.

The principles discussed below include a set of criteria represented using the notation “x.y.z”, where each element has a specific meaning:

- x: principle (1. Perceivable, 2. Operable, 3. Understandable, and 4. Robust).
- y: guideline.
- z: success criterion (each guideline contains several success criteria that describe the specific conditions required to satisfy that guideline).

This numbering system provides a hierarchical structure for addressing different aspects of web accessibility.

# PRINCIPLE 1: PERCEIVABLE DESIGN

Information and user interface components must be presented to users in ways they can perceive and understand. This principle focuses on ensuring that information and user interface operations are made available in forms that users can perceive.

To achieve this, **four guidelines** are established:

- **Text Alternatives:** Provide alternative text for non-text content (large print, Braille, sign language, symbols, or simpler language).
- **Time-Based Media:** Provide alternatives for audio and video content.
- **Adaptable:** Create content that can be presented in different ways without losing information or structure.
- **Distinguishable:** Make it easier for users to see and hear content, including separating foreground information from background content.

# PRACTICAL CASES

Based on specific guidelines and techniques used to evaluate the Perceivable Design Principle:

**a) Color Contrast (Success Criterion 1.4.3).**

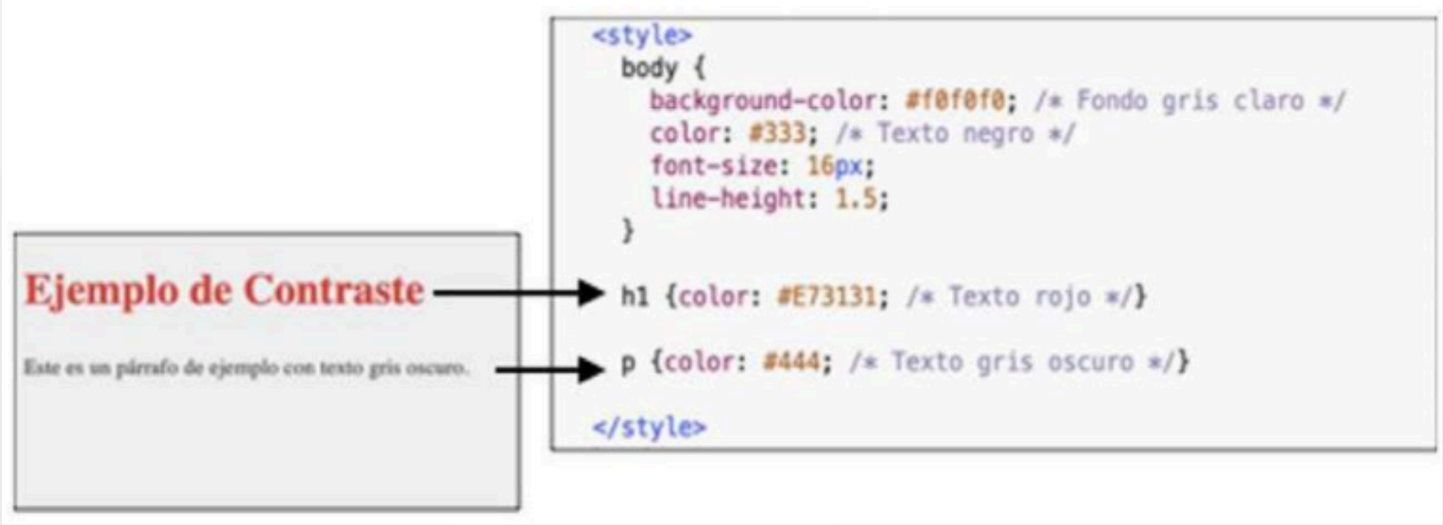
- Text (and images of text) must have a contrast ratio of at least **4.5:1** against its background, except for large text, incidental text, and text embedded in images. In other words, there must be sufficient contrast between the text and the background to ensure that the text is legible.

- To verify compliance with this criterion, as well as others, color contrast checking tools can be used. In Figure, the dark gray color of the text line provides better contrast with the background color than the title color does (as will be confirmed later using accessibility evaluation tools).

# PRACTICAL CASES (II)

Based on specific guidelines and techniques used to evaluate the Perceivable Design Principle:

**a) Color Contrast (Success Criterion 1.4.3) (II)**



# PRACTICAL CASES (III)

**b) Text Adaptability (Success Criterion 1.4.4).**

This criterion allows users to adjust text size without losing information or functionality. It is directly related to responsive design. It can be implemented using the **@media** rule to apply additional styles when the screen width is **600 px or less**, making it possible to adjust text size for smaller devices.

**c) Text Alternatives for Multimedia (Success Criteria 1.2.1 and 1.2.2).**

Text alternatives should be provided for non-text content, such as images and multimedia. This can be achieved by using the **alt attribute**, followed by a textual description of the image or video.

# PRACTICAL CASES (IV)

**d) Content Adaptability (Success Criterion 1.4.10).**

Content must be adaptable and capable of being presented in different ways, such as on mobile devices, without losing information or functionality. This criterion is also related to the @media rule, which allows content to adapt to different screen sizes and mobile devices.

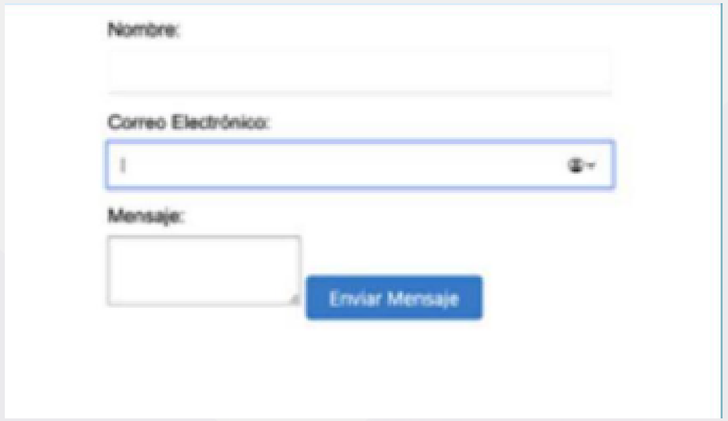
**e) Contrast and Presentation (Success Criterion 1.4.11).**

A mechanism must be provided to pause, stop, hide, or adjust the speed of content that moves, blinks, flashes, or automatically updates for more than five seconds. For example, if a CSS animation is moving an element across the screen, a method should be included to stop it, such as a button.

# PRACTICAL CASES (V)

**f) Component Recognition (Criterion 1.3.1).**

Interactive elements, such as forms, must have appropriate labels and associations so that users can understand and complete them. In this example, the form field is labeled using the <label> tag, which is associated with the corresponding field through the for and id attributes. This allows the input field to be selected when the user clicks on the "Email Address" label.



# PRACTICAL CASES (VI)

**g) Use of Color Is Not the Only Means of Communication (Success Criterion 1.4.12).**

This criterion aims to ensure that information is not presented in a way that depends exclusively on color perception. To achieve this, websites should be designed so that content remains understandable without relying solely on color, for example, by using styles, text, and structural elements to ensure information is accessible to all users regardless of their ability to perceive color.

**h) Meaningful Sequence and Structure of Information (Success Criterion 1.3.3).**

Techniques should be used to help users understand and interpret information. For example, using headings such as <h1> and <h2> to organize content and make its structure clear.

# PRINCIPLE 2: OPERABLE DESIGN

User interface components and navigation must be operable. This principle focuses on ensuring that users can effectively interact with and operate both the interface and its content.

To achieve this, **four guidelines** are established:

- **Keyboard Accessible:** All functionality should be available through a keyboard.
- **Enough Time:** Users should be given sufficient time to read and use the content.
- **Seizures and Physical Reactions:** Content should not be designed in a way that could trigger epileptic seizures.
- **Navigable:** Provide ways to help users navigate, find content, and determine where they are within the website.

# PRACTICAL CASES

**a) Keyboard Accessibility (Success Criterion 2.1.1).**

All important functions and interactive areas of a website must be accessible and usable through the keyboard. To ensure this, keyboard navigation testing should be carried out without using a mouse. For example, if there is a button, it should be accessible and functional using both the mouse and the keyboard. Particular attention should be paid to events associated with the Enter key.

**b) Enough Time (Success Criterion 2.2.1).**

Users must be given sufficient time to read and use the content. Therefore, content should not change or update automatically without providing an option to pause, stop, or adjust the timing. JavaScript functions such as setInterval() can be used to manage timed content, provided users are given appropriate controls.

# PRACTICAL CASES (II)

**c) Flashing and Blinking Content (Success Criterion 2.3.1).**

It is necessary to incorporate measures that prevent the use of elements that flash or blink automatically, as these could trigger seizures. Therefore, a method should be provided to disable or adjust the blinking effect, for example, by using JavaScript.

```
<script>
  var parpadeo = document.getElementById('parpadeo');

  function detenerParpadeo() {
    parpadeo.style.animation = 'none';
  }
</script>
```

**d) Intuitive Navigation (Success Criterion 2.4.2).**

The structure of a website should be intuitive and predictable. To achieve this, the use of clear headings, labels, and links is recommended, helping users quickly understand the site's organization and navigate efficiently.

# PRACTICAL CASES (III)

**e) Distractions (Success Criteria 2.2.2 and 2.3.2).**

Distractions on a website should be minimized. For example, content that blinks, flashes, or changes automatically should be avoided unless users are given the ability to control, pause, or stop it.

**f) Search-Based Navigation (Success Criterion 2.4.5).**

It is advisable to provide a search function to help users find specific content efficiently. This can be implemented by including a small form with a text input field and a search button. This feature provides a quick and convenient way for users to locate information within the website.

# PRACTICAL CASES (IV)

**g) Consistent Navigation (Success Criterion 3.2.3).**

The website should maintain consistency in its navigation and design so that users can predict how the interface will behave. For example, consistency can be achieved by using similar styles for the header and navigation bar, as well as by placing common elements in predictable locations throughout the site.

# PRINCIPLE 3: UNDERSTANDABLE DESIGN

Information and user operations must be understandable.

To achieve this, three guidelines are established:

- **Readable:** Make text content readable and easy to understand.
- **Predictable:** Ensure that web pages have a predictable appearance and behavior.
- **Input Assistance:** Help users avoid and correct errors when entering data.

# PRACTICAL CASES

**a) Clear Language (Success Criterion 3.1.2).**

Clear language should be used throughout the website, avoiding unnecessary technical jargon or complex expressions. Information should be easy to understand and accessible to a broad audience.

**b) Predictability (Success Criterion 3.2.1).**

The design of the elements within a website should ensure that navigation and interaction are predictable. In other words, users should be able to anticipate how interface elements will behave and what will happen when they interact with them.

# PRACTICAL CASES (II)

**c) Data Input (Criterion 3.3.1).**

It is advisable to provide suggestions and assistance, for example, by including the correct format for information or instructions about the type of data expected in a field. This helps users complete input fields, indicating, when necessary, the use of a specific format, among other details.

**d) Error Identification (Criterion 3.3.3).**

Users should be informed clearly and understandably about any errors in forms or other types of interactions. Therefore, if users make mistakes while completing forms or interacting with the interface, the reason for the error should be indicated (Figure).

# PRACTICAL CASES (III)

**e) Consistency (Success Criterion 3.2.2).**

Consistency should be maintained throughout the website’s design, especially in the presentation of content, so that users can understand and predict how the interface will behave.

**f) Contextual Help (Success Criterion 3.3.2).**

As discussed in previous criteria, it is advisable to provide contextual help to assist users in understanding information or specific functionality.

# PRACTICAL CASES (IV)

**g) User Input (Success Criterion 3.3.4).**

Users should be able to correct or reverse unintended actions and be informed about how to do so. For example, when incorrect data is entered into a form, a script should prevent the form from being submitted if the data is invalid and display an alert message to the user. This allows users to correct the information or undo the submission if it was performed unintentionally.

# PRINCIPLE 4: ROBUST DESIGN

Content must be robust enough to be interpreted by a wide variety of user agents, including **assistive technologies**. To achieve this, the following guideline is established:

- **Compatible:** Maximize compatibility with current and future user agents, including assistive technologies.

# PRACTICAL CASES

**a) Compatibility with Assistive Technologies (Success Criterion 4.1.2).**

Content should be compatible with assistive technologies such as screen readers and voice browsers. For example, the aria-label attribute can be used on a button to provide additional descriptive text for users of assistive technologies, such as screen readers.

**b) Browser Compatibility (Success Criterion 4.1.1).**

The website should, as far as possible, be compatible with a wide range of web browsers.

# PRACTICAL CASES (II)

**c) Detection of Changes (Success Criterion 4.1.3).**

Changes in content or in the state of the user interface should be detectable and communicated to users, which is especially important when assistive technologies are being used. For example, aria-live="polite" can be used to notify assistive technologies about updates or changes in content.

# CONFORMANCE REQUIREMENTS

A web page must satisfy all of the following conformance requirements in order to comply with WCAG 2.0:

**1. Conformance Level.**

Verifiable statements about web content that are used to assess whether a website meets accessibility requirements. Three levels of conformance are distinguished, defining the degree of accessibility; one of these levels must be fully satisfied.

**Level A:** To achieve Level A conformance (the minimum level), the website satisfies all Level A success criteria or provides a conforming alternative version. A total of 25 criteria must be met.

# CONFORMANCE REQUIREMENTS (II)

A web page must satisfy all of the following conformance requirements in order to comply with WCAG 2.0:

**1. Conformance Level.**

Levels (II)

- **Level AA:** To achieve Level AA conformance, the website satisfies all Level A and Level AA success criteria, or provides a conforming alternative version that meets Level AA requirements. A total of 13 criteria must be met.

- **Level AAA:** To achieve Level AAA conformance, the website satisfies all Level A, AA, and AAA success criteria, or provides a conforming alternative version that meets Level AAA requirements. A total of 23 criteria must be met.

# CONFORMANCE REQUIREMENTS (III)

**2. Complete Pages.**

Conformance applies to complete web pages. It cannot be claimed if any part of a page is excluded.

**3. Complete Processes.**

When a web page is part of a series of pages that make up a process, all pages within that process must conform to the specified level or a higher one.

# CONFORMANCE REQUIREMENTS (IV)

**4. Use of Accessibility-Supported Technologies Only.**

Meeting the success criteria must depend only on technologies that are compatible with accessibility requirements.

**5. Non-Interference.**

If technologies are used in a way that is not accessibility-supported, or if they do not meet conformance requirements, they must not prevent users from accessing the content in the rest of the page.

# TYPES OF DISABILITIES AND HOW TO IMPLEMENT ACCESSIBILITY ON THE WEB

There are various types of users depending on their cognitive, technological, or cultural abilities. Another important aspect to consider is disability.

The most significant disabilities related to web interface design are visual, motor, and hearing disabilities.

Different types of technologies applied to accessibility will be studied.

# VISUAL DISABILITY

A visual disability is defined as the difficulty some people experience in participating in everyday activities due to the interaction between a specific impairment related to the **reduction** or **loss of visual functions** and the barriers present in the environment in which the person operates.

- **Blindness:** there is a complete limitation of visual function.
- **Visual impairment:** the limitation is not total

The **measures** that need to be taken in each case are different. For example, for people with total blindness, a possible solution would be to design a version of the interface with larger font sizes than those used in the original version.

# ACCESS BARRIERS FOR PEOPLE WITH VISUAL DISABILITIES TO AVOID

1. **Inability to access content or operate the web application using only the computer keyboard.**

It is necessary to consider its **full functionality and usability through the keyboard**. One common example is submitting a form through a button that can only be activated with the mouse, or a menu that only expands when the mouse pointer hovers over it. In these cases, users should be able to assign **keyboard shortcuts** to any link or form control to allow keyboard navigation.

It is important to ensure that the creation of these shortcuts does not conflict with shortcuts already defined by the operating system. It is recommended to use keyboard shortcuts only for elements that are accessed very frequently on the page or are difficult to locate.

# ACCESS BARRIERS FOR PEOPLE WITH VISUAL DISABILITIES TO AVOID (II)

2. **Absence of alternative text for non-text elements.**

3. **Complex forms and data tables that are difficult to interpret correctly.**

When creating forms, it is especially important to take certain accessibility aspects into account in order to ensure that the **content can be correctly interpreted by screen reader** users.

# SCREEN READERS

One of the tools used by people with visual disabilities is a screen reader, which is software that “reads” or explains the content displayed on the screen.

In the case of **forms**, the software must be able identify the **type of control** being used, indicate its **status**, and announce the **label** associated with that control.

Some of the most widely used screen readers today are NVDA, JAWS, VoiceOver (Apple devices), and TalkBack (Android devices). They allow to access and navigate web content through speech synthesis and, in many cases, Braille displays.

For those who have a **visual impairment**, there are also **screen magnifiers**, which are programs that allow text and images to be enlarged. In addition, there are **specialized browsers** that can read web pages aloud and recognize the user's voice for navigation purposes.

# ACCESS BARRIERS FOR PEOPLE WITH VISUAL DISABILITIES TO AVOID (III)

4. **Improper use of structural elements on web pages or a lack of structure in their content:** missing section headings, list definitions, groupings of controls, etc. The way the content of a web page is structured is a fundamental part of its accessibility. It is very important to correctly identify basic structural elements such as headings, lists, paragraphs, data tables, and so on.

5. **Websites with poor color contrast or information conveyed solely through color.**

6. **CAPTCHAs that do not provide an accessible alternative.** Visually distorted text images used as control mechanisms to distinguish a human user from a computer program, should provide different alternative methods of accessing the information, adapted to different sensory abilities.

# WHY SHOULD SPECIAL ATTENTION BE PAID TO HEADINGS?

Their function is to briefly describe the topic of the section they introduce.

On a web page, it is common to find section titles that visually stand out from the rest of the content or areas of content that are visually separated to address different topics. These are clear situations in which section headings (h1 to h6) should be used.

Thanks to the correct use of these structural elements, assistive technologies can make navigation easier for users who access the content of a web page or document in a linear way. This is especially important for blind users, for whom the ability to navigate through headings is extremely useful.

# HEARING DISABILITY

The total or partial loss of the ability to perceive sound, which is assessed according to the degree of hearing loss in each ear.

- **Hard-of-hearing individuals**: they have a partial hearing impairment and therefore retain some hearing ability that may be improved through the use of **hearing aids**.

- **Deaf individuals**: they have a total hearing impairment.

The main accessibility barrier in this case is the increasing use of **audiovisual elements** on the web (videos, animations, sounds, etc.). To make these resources accessible to people with hearing disabilities, they should provide a **text-based alternative**.

# MOTOR DISABILITY

It is a condition in which a person experiences a long-term, and often chronic, impairment of the musculoskeletal system that results in limitations in daily activities related to mobility. This category also includes disorders that may cause partial or total deterioration of motor skills, affecting either the upper body, the lower body, or both.

In terms of website design, some users may encounter difficulties in common computer tasks, such as:

- using a mouse
- moving a pointer
- using a touchscreen
- pressing two keys simultaneously
- holding down a key.

Users may even be unable to use a keyboard and, therefore, may not be able to enter data.

# STICKY KEYS

If it is necessary to press several keys at the same time and using both hands is not possible, **operating systems** usually include software that provides the so-called **Sticky Keys feature**.

This system works by creating key patterns in which repeated presses of a particular key are interpreted as key combinations

# ACCESS BARRIERS FOR PEOPLE WITH MOTOR DISABILITIES TO AVOID

1. **Inability to interact properly with a web page using the keyboard or other input devices.**

To facilitate interaction with web pages, it is recommended to use **specialized input devices** designed for computer use in general, and web navigation in particular. Examples include:

- Keyboards with large keys
- Conventional keyboards with keyguards
- Large trackball mouse
- Button-operated mouse, among others.

# ACCESS BARRIERS FOR PEOPLE WITH MOTOR DISABILITIES TO AVOID

2. **Graphic links and other interactive elements that are not correctly labeled and are inaccessible to voice recognition software.**

Once again, as in the case of visual disabilities with screen readers, voice recognition systems are especially important in this context, as they allow users to perform the desired actions through voice commands.

For these systems to function properly, interface elements must be designed so that they can be easily activated using this type of technology.

# ASSISTIVE TECHNOLOGIES

As mentioned in the previous point, the development of assistive technologies has greatly improved website accessibility. In this case, it is s**oftware and input devices that recognize our actions.** For example:

- **Speech recognition software.** Designed for situations in which a user cannot use a keyboard or mouse.

- **Trackball.** A device similar to a mouse, but it does not require physical movement across a surface. Instead, a large ball is rotated to move the pointer around the screen.

- **Webcams.** Through facial recognition technology, webcams can translate head or eye movements into mouse movements on the screen.

# ARIA ATTRIBUTE

ARIA (Accessible Rich Internet Applications) consists of a set of HTML attributes used to improve the accessibility of web applications, especially those that rely on interactive technologies such as JavaScript. These attributes are particularly useful for providing information to assistive technologies, such as screen readers, helping people with disabilities better understand and interact with the user interface. Some of the **most commonly used ARIA attributes** are described below:

- **aria-label.** Adds a text label that describes the function of an element.

```
<button aria-label="Cerrar ventana" onclick="cerrarVenta-
na()">Cerrar</button>
```

- **aria-labelledby.** Associates an element with another element on the page that acts as its label.

```
<h2 id="titulo-ventana">Ventana de Configuración</h2>
<div role="dialog" aria-labelledby="titulo-ventana">
      <!-- Contenido de la ventana de configuración -->
</div>
```

# ARIA ATTRIBUTE (II)

Some of the **most commonly used ARIA attributes** (II):

- **aria-describedby.** Associates an element with another element on the page that provides a more detailed description.

```
<input type="text" aria-describedby="instrucciones-input">
<p id="instrucciones-input">Ingrese su nombre completo</p>
```

- **aria-hidden.** Determines whether an element and its descendants will be visible to assistive technologies by setting its value to true or false.

```
<div aria-hidden="true">Contenido no visible para tecnolo-
gías de asistencia</div>
```

# ARIA ATTRIBUTE (III)

Some of the **most commonly used ARIA attributes** (III):

- **aria-live.** Used to determine whether content updates should be announced immediately to the user or whether the user should wait for an update. This attribute can take several **values to specify how real-time content updates should be handled**:

  - off: disables automatic announcement of updates.
  - assertive: causes content updates to be announced immediately, interrupting any other task that is currently being read.
  - polite: causes content updates to be announced after the user has finished the current interaction.

```
<div aria-live="assertive">¡Nuevo mensaje recibido!</div>
```

# ACCESSIBILITY EVALUATION TOOLS

The process of evaluating the accessibility of a website is divided into **two phases**.

First, an **automatic analysis** should be carried out to detect accessibility issues.
Second, a **manual evaluation** must be performed to identify problems that cannot be detected during the first phase.

Some of the most widely recognized **automated tools currently available** include:

- **W3C HTML Validator**. Performs a validation of a website's grammar, checking whether HTML documents conform to the standards and specifications defined by the W3C (World Wide Web Consortium). It allows the inspection of files stored locally or accessed through a URL.

# ACCESSIBILITY EVALUATION TOOLS (II)

- **Lighthouse.** An open-source auditing tool developed by Google that provides detailed reports on performance, accessibility, best practices, SEO, and other aspects of a website. It is easy to use and is available within Chrome Developer Tools. Users simply need to select the "Lighthouse" option and load the website they wish to audit. It can analyze both HTTP and HTTPS websites.

- **WAVE (Web Accessibility Evaluation Tool)**. An online tool that evaluates the accessibility of a website and generates detailed reports identifying specific accessibility issues.

- **W3C CSS Validator.** Validates the grammar of CSS style sheets and verifies the correctness of HTML documents that use style sheets.

# ACCESSIBILITY EVALUATION TOOLS (III)

- **WCAG Contrast Checker.** This tool allows designers and developers to verify the color contrast between two colors, ensuring compliance with the color contrast requirements defined by the WCAG (Web Content Accessibility Guidelines).

- **Axe DevTools (Chrome and Firefox Extension)**. An extension that integrates directly into the browser's developer tools, allowing real-time accessibility audits to be performed while developing or testing a website.

- **Color Contrast Analyzer.** A desktop application that enables users to analyze color contrast directly within a user interface. It allows specific colors to be selected for evaluation and provides a summary indicating whether they meet the relevant WCAG accessibility criteria

# ACCESSIBILITY EVALUATION TOOLS (IV)

- **Contrast Ratio.** An online tool used to evaluate the contrast ratio between two colors according to the formula established by the Web Content Accessibility Guidelines (WCAG). Color contrast is a crucial factor in ensuring web accessibility, particularly for people with visual impairments.

# REGULATIONS

**Royal Decree 1112/2018** provides the **legal framework** that defines and establishes the accessibility and usability requirements for websites and mobile applications of the public sector in Spain.

Today, the development of websites and mobile applications for Public Administration is becoming increasingly widespread as part of an essential digital transformation process aimed at creating new strategic assets.

This regulation is key to any development applied to websites and mobile applications belonging to public sector organizations, including government bodies, local administrations, and other entities linked to the public sector.

# REGULATIONS (II)

The regulation establishes universal accessibility principles that websites and mobile applications must comply with in order to guarantee access to information and services for all people, regardless of their abilities or disabilities. The accessibility requirements are defined at **three conformance levels**, identified as A, AA, and AAA, following the W3C Web Content Accessibility Guidelines (WCAG) 2.1.

Likewise, **UNE 301549:2019** is a Spanish technical standard that establishes accessibility and usability requirements for web content, software, and mobile devices, particularly those used in electronic public services.

It is fully aligned with the Web Content Accessibility Guidelines (WCAG) of the World Wide Web Consortium (W3C). The standard addresses aspects such as visual perception, hearing, mobility, understandability, and operability.

# USABILITY ANALYSIS AND VERIFICATION

It is essential to be able to analyze and verify the level of usability of a given web portal based on the objectives and needs of its users.

This information makes it possible to identify potential improvements or modifications that can enhance the website’s usability.

The process of usability analysis and verification is usually carried out during the evaluation phase of a web portal project, before it is deployed into production.

Various methods are commonly used for this type of assessment. The following slides describe some of the methods most frequently applied in web application development.

# INSPECTION METHOD: HEURISTIC EVALUATION

This method is carried out by usability experts who thoroughly analyze a website in order to identify potential issues that need to be corrected. The basis of this method lies in the evaluators' own experience, as well as in established best-practice guidelines used to detect principles related to usability. Some of these principles include:

1. **Compliance with accessibility guidelines.**
2. **Use of the same language and terminology between the website and the user.**
3. **Providing feedback to the user about what the system is doing**. Informing users about what is happening during a process. A typical example is when a user attempts to access audiovisual content offered by the website and the portal displays a message indicating that the content is currently loading into the buffer.

# INSPECTION METHOD: HEURISTIC EVALUATION (II)

Principles (II):

4. **Promoting user control over the website.** It is important that users can, for example, undo an action if they have made a mistake. Likewise, the website should assist and guide users in resolving errors, such as when completing a form field incorrectly.

5. **Including help documentation that users can consult whenever needed.**

6. **Improve the efficiency of the web portal.** In this way, the website will be user-friendly and easy to use for users who have never used it before, while also providing shortcuts for advanced users.

# INSPECTION METHOD: HEURISTIC EVALUATION (III)

Principles (III):

7. **Appropriate organization of the information on the website.**

8. **Inclusion of multimedia elements on the website and adaptation of these elements to the site's theme.**

9. **Appropriate content quality in terms of language use and writing style.**

✅ The main advantage of this method compared to others is that it can be developed in a simple, fast, and effective way within a short period of time, although its cost may be higher.

# USER TESTING METHOD

- This approach is based on analyzing a website through a group of real users, allowing those users to detect usability problems or propose improvement options.

- This method can be implemented once the website is in production, but it can also be carried out during the design and implementation phases. This is a very important aspect, since **correcting problems during the web design stages is less costly than doing so during the production phase**.

- This method should not be considered the opposite of the inspection method. Instead, both methods complement each other in the analysis of usability within a web environment.

# USER TESTING METHOD (II)

- ✅ The main advantage of the user testing method is that the **results are more reliable** and allow for a **deeper identification of design errors** related to the site's usability.

- However, it is important to consider the **high cost** involved in its implementation, since it requires several users and the provision of a suitable area in which they can perform their tasks, as well as the appropriate tools needed to carry them out.

# ANALYSIS AND VERIFICATION OF THE WEBSITE

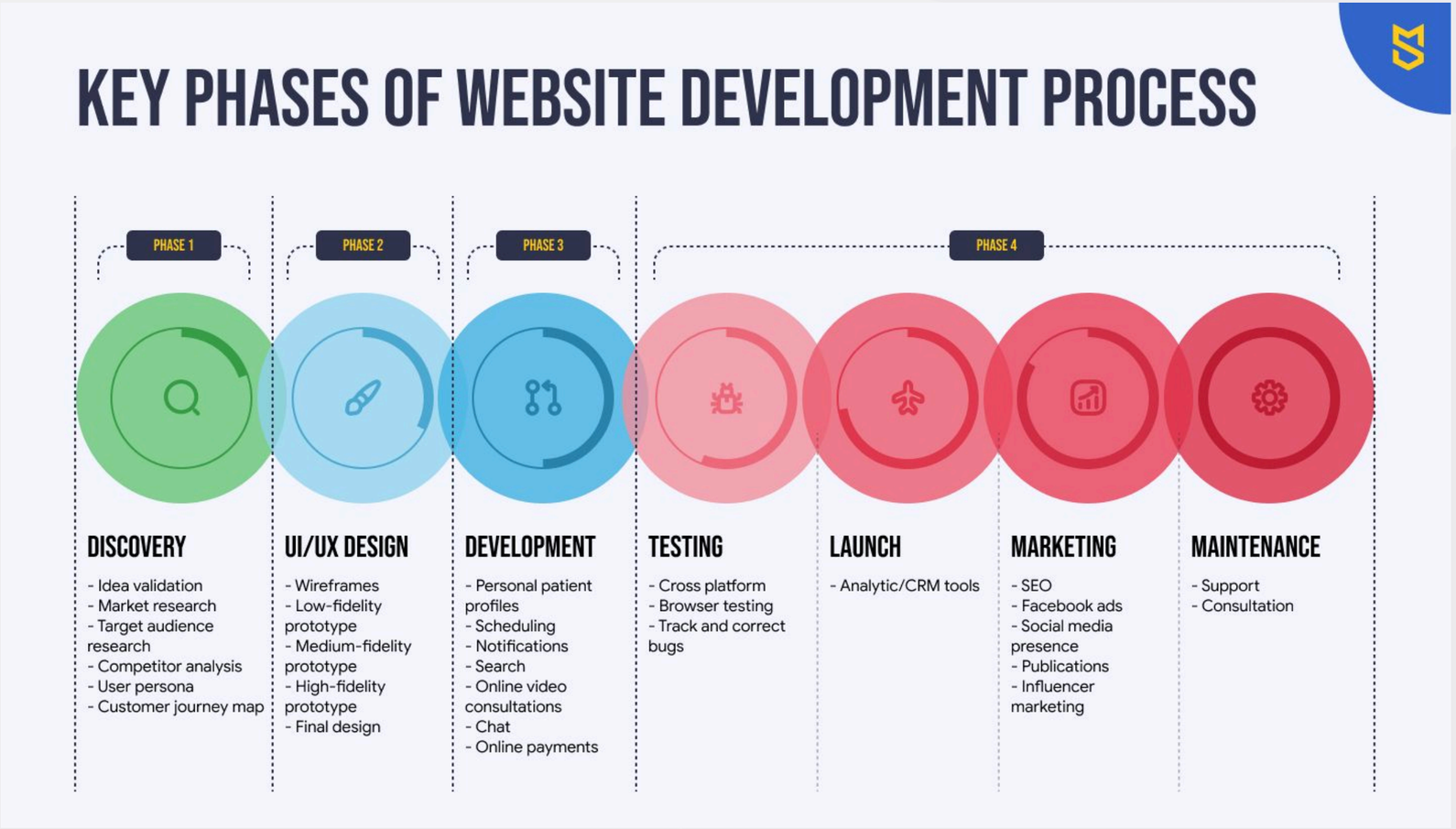
- When implementing a particular web design (as with any other project), it is necessary to establish a series of phases or processes to facilitate the achievement of the objective.

- That is, a series of procedures, techniques, and specific methods must be applied to accomplish the intended goal.

# ANALYSIS AND VERIFICATION OF THE WEBSITE (II)

**Figura 11.9**
Diagrama y distribución de las fases de análisis y planificación de un sitio web.

# ANALYSIS AND VERIFICATION OF THE WEBSITE (III)



# ANALYSIS AND VERIFICATION OF THE WEBSITE (IV)

- The analysis and planning phases of a website must be adapted to the objectives to be achieved and to the needs that the web portal is intended to address.

- In cases where the goal is to create a website that ensures usability, accessibility, and navigability, it is possible to speak of **user-centered web design.**

- All phases should be focused on achieving this objective, which makes it necessary to involve users from the **planning stage** in order to guarantee their satisfaction and address their needs and requirements.

- It is also highly important to involve users during the **evaluation phase**, so that their user experiences can be assessed and any necessary improvements can be implemented.

# PLANNING PHASE

- Every successful project, regardless of its purpose, needs to be planned before work begins. Web design projects are no different.
- During this phase, the main goals of the website should be defined, along with the methods and resources that will be used to achieve them.
- When planning a website, it is important to consider factors such as the professional and technical resources required, hosting options, costs, and other practical aspects.
- It is also necessary to decide what content the site will include and how that content will be organized to create a positive user experience.

# PLANNING PHASE (II)

- Another key task during this stage is gathering information about the **target users**.
- This can be done through surveys, interviews, meetings, or other research methods.
- The objective is to better understand users' needs, goals, and behaviors so that the website can be designed to meet their needs, objectives, behaviors, and preferences.

# DESIGN PHASE

- During this phase, the **appearance** and **functionality** of the website will be developed, always taking **usability** into account, as well as all the information gathered during the planning phase.

**Key Point:**

- It is essential for the designer to **consider the different types of users** who may visit the website.
- Understanding their needs, limitations, and abilities helps ensure that the site is adapted to them.
- As a result, the website will not only be more user-friendly but also more accessible and easier to navigate.

# DESIGN PHASE (II)

- When designing a website, it is highly recommended to create a **content and site organization** plan beforehand, identifying the different menus and pages that will be included. This helps define the overall structure of the website and prevents potential issues later in the development process.

- It is also advisable to **properly document all design decisions and actions taken**. This makes it easier for other designers or developers to understand the project if they need to work on it in the future.

- Regarding the design of the website itself, accessibility should be a key consideration. This involves establishing criteria such as a**ppropriate page margins**, **suitable image sizes and resolutions**, and **sufficient contrast between text and background**, among other factors.

# IMPLEMENTATION PHASE

- During this phase, the design created in the previous stage is put into practice, resulting in the first working versions of the website.
- At this point, it is important to have carefully selected the web technologies, programming languages, and development platforms that will be used.
- It is also highly recommended **to create early versions or prototypes of the website**.
- These prototypes can be tested and reviewed to evaluate aspects such as user experience, usability, and accessibility before the final version is completed.

# EVALUATION PHASE

- This is one of the most important stages of the process.
- During this phase, a decision is made on whether the website is ready to go live or if certain aspects need to be reviewed and improved before release.
- Based on the results of the evaluation, the project may move forward to production, return to the design phase for adjustments, or even go back to an earlier stage if significant changes are required.
- There are several methods commonly used to evaluate website usability, as discussed previously.

# PRODUCTION PHASE

- This is the stage where the website is launched and made available to its users, once its operation has been tested and verified during the previous phases of the process.

- During this stage, it is also important to promote the website effectively among its target audience in order to increase awareness and attract more users.

# MAINTENANCE AND MONITORING PHASE

- This is the final phase of the process, although it should be considered just as important as the website itself. Web technologies are constantly evolving, and users' needs and expectations change over time. For this reason, it is essential to keep the website updated and continuously improve it.

- The website should be monitored regularly to identify and correct any issues and to analyze how users interact with the site. This information can be collected through logs, analytics tools, and other monitoring methods.

- To support continuous improvement, it can also be useful to provide ways for users to share their opinions, suggestions, and feedback about the website, helping to identify opportunities for future enhancements.

# SEARCH ENGINE OPTIMIZATION (SEO)

- Search Engine Optimization (SEO) is a key aspect of web design, as it plays an important role in determining how visible and successful a website is in search engine results.

- SEO is still important, but modern SEO is less about manipulating search engines and more about creating useful content and a good user experience. AI has not replaced SEO; it has made quality content and website structure even more important.

- Google is still relevant for local businesses: https://www.brightlocal.com/research/consumer-search-behavior-channels/

# KEY PRINCIPLES AND TECHNIQUES

Web interface design should incorporate a variety of practices and techniques to improve a website's ranking in search engine results:

- **Organized Structure.** A website should be designed to be user-friendly for both visitors and search engines. It should be easy for users to navigate while also allowing search engines to efficiently crawl and index its content.

- **Content Optimization.** Website content should be relevant, useful, and high quality, providing clear answers to users' questions. It should also target important keywords related to the business or industry, helping to improve search engine visibility and SEO performance. In addition to helping a website rank in traditional search results, high-quality content increases the likelihood of being referenced by AI-powered search tools and search engine AI summaries.

# KEY PRINCIPLES AND TECHNIQUES (II)

- **Responsive Design.** Responsive design remains essential not only because **most users browse on mobile devices**, but also because search engines and AI-powered search systems prioritize websites that provide a consistent, high-quality experience across different devices and screen sizes.

- **Performance and Speed.** Search engines generally favor websites that load quickly because they provide a better user experience. It is important to optimize website performance through techniques such as image compression, browser caching, and minimizing HTML, CSS, and JavaScript code.

- **Internal Link Structure Optimization.** A website should have a clear and well-organized internal linking structure that makes navigation easier for users and helps search engines understand the relationships between different pages. Well-designed internal links can improve indexing and distribute page authority more effectively throughout the site.

# ON-PAGE AND OFF-PAGE SEO

**On-page SEO** focuses on all the actions and strategies implemented within a website to improve its visibility in search engines and increase organic traffic. This approach includes a variety of techniques, such as keyword optimization and proper website configuration.

- Improving website functionality.
- Optimizing all website content.
- Creating high-quality, user-friendly URLs.
- Making content easier for search engines to index.
- Enhancing search engine understanding of the website.
- Improving website speed and performance.

# ON-PAGE AND OFF-PAGE SEO (II)

**Off-page SEO** focuses on external factors that influence a website's search engine ranking. It involves activities carried out outside the website that help improve its visibility and authority in search results, including:

- Building external links (backlinks) to attract traffic from other websites.
- Creating high-quality content for social media platforms.
- Leveraging social media influence to connect with and engage the target audience.

While on-page and off-page SEO remain important, modern SEO also considers how content may be discovered and referenced by AI-powered search tools. High-quality, trustworthy, and well-structured content is increasingly important for visibility in both traditional search results and AI-generated answers.

# ¿ANY QUESTION?